

Environmental Disturbance Torques on a 3U CubeSat

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Abstract

A reaction-wheel attitude control system must reject the small but persistent external torques that act on a spacecraft in low Earth orbit. This note derives, from first principles, the four dominant environmental disturbance torques for our 3U CubeSat reference mission (a 500 km Sun-synchronous orbit): the gravity-gradient torque, aerodynamic drag torque, solar radiation pressure torque, and residual magnetic torque. For each, we state the physical origin, derive the governing equation, and evaluate the order of magnitude for our platform. All four are of order 10^{-9} N m to 10^{-6} N m; over an orbit they pump momentum into the wheels, which is periodically removed by magnetorquer desaturation.

1 Reference Parameters

All numerical estimates use the reference mission and CAD-derived properties in Table 1.

Table 1: Reference orbit and spacecraft parameters.

Quantity	Symbol	Value
Orbit altitude	h	500 km
Orbit radius	r	6878 km
Orbital speed	v	7613 m/s
Mean motion	n	1.107×10^{-3} rad/s
Max transverse inertia	$I_{\max}$	1.405×10^{-2} kg m ²
Min inertia (spin axis)	$I_{\min}$	2.889×10^{-3} kg m ²
Long-face area	A	0.034 m ²
Centre-of-pressure offset	r_{cp}	0.02 m
Earth magnetic field (LEO)	B	2×10^{-5} T to 5×10^{-5} T

The body-to-inertial rotation matrix built from the attitude quaternion $\mathbf{q}$ is written $\mathbf{R}(\mathbf{q})$; its transpose $\mathbf{R}(\mathbf{q})^\top$ maps an inertial vector into the body frame. All torques below are expressed in the body frame, because that is where Euler’s equation is integrated.

2 Gravity-Gradient Torque

Physical origin

Earth’s gravity is stronger on the near side of the spacecraft than the far side. For an extended body this differential pull produces a torque that tends to align the axis of minimum inertia with the local vertical.

Derivation

Model the body as a collection of point masses m_i at positions $\mathbf{r}_i^c$ relative to the centre of mass (CoM). The gravitational force on each is expanded to first order about the CoM,

$$\mathbf{F}_i \approx m_i [\mathbf{g}(\mathbf{r}^{c0}) + \mathbf{G}(\mathbf{r}^{c0}) \mathbf{r}_i^c], \quad (1)$$

where $\mathbf{G} = \partial \mathbf{g} / \partial \mathbf{r}$ is the gravity-gradient tensor at the CoM. The total torque about the CoM is $\mathbf{L} = \sum_i \mathbf{r}_i^c \times \mathbf{F}_i$. The uniform part $(\sum_i m_i \mathbf{r}_i^c) \times \mathbf{g}$ vanishes identically, because $\sum_i m_i \mathbf{r}_i^c = \mathbf{0}$ *by the definition of the centre of mass*. Only the gradient term survives; carrying out the sum (which turns the mass distribution into the inertia tensor $\mathbf{J}$) and specialising to a spherical Earth gives the standard result

$$\boxed{\boldsymbol{\tau}_{gg} = 3n^2 \hat{\mathbf{o}} \times (\mathbf{J} \hat{\mathbf{o}})} \quad (2)$$

where $\hat{\mathbf{o}} = \mathbf{R}(\mathbf{q})^\top \hat{\mathbf{o}}_{\text{inertial}}$ is the nadir (satellite $\rightarrow$ Earth) unit vector expressed in the body frame, and n is the orbital mean motion.

Key properties

- The torque *vanishes* when a principal axis points at nadir, since then $\hat{\mathbf{o}}$ is an eigenvector of $\mathbf{J}$ and $\hat{\mathbf{o}} \times \mathbf{J} \hat{\mathbf{o}} = \mathbf{0}$. This is the basis of passive gravity-gradient stabilisation.
- It is attitude-dependent: it must be recomputed every integration step.

Order of magnitude

The worst-case magnitude is

$$\tau_{gg}^{\max} = \frac{3}{2} n^2 |I_{\max} - I_{\min}| = \frac{3}{2} (1.107 \times 10^{-3})^2 (1.405 \times 10^{-2} - 2.889 \times 10^{-3}) \approx 2.1 \times 10^{-8} \text{ N m.} \quad (3)$$

3 Aerodynamic Drag Torque

Physical origin

Even at 500 km the residual atmosphere exerts a drag force on the spacecraft. If the centre of pressure (CP) does not coincide with the CoM, this force produces a torque.

Derivation

The drag force opposes the velocity direction $\hat{\mathbf{v}}$ and has magnitude

$$\mathbf{F}_{\text{drag}} = -\frac{1}{2} \rho v^2 C_d A \hat{\mathbf{v}}, \quad (4)$$

with ρ the local atmospheric density, v the orbital speed, C_d the drag coefficient (typically 2.2 for free-molecular flow), and A the projected frontal area. Expressed in the body frame and applied at the CP offset $\mathbf{r}_{cp}$ from the CoM, the torque is

$$\boxed{\boldsymbol{\tau}_{\text{aero}} = \mathbf{r}_{cp} \times (\mathbf{R}(\mathbf{q})^\top \mathbf{F}_{\text{drag}})} \quad (5)$$

Order of magnitude

With $\rho \approx 5 \times 10^{-13} \text{ kg/m}^3$ (moderate solar activity), $v = 7613 \text{ m/s}$, $C_d = 2.2$, $A = 0.034 \text{ m}^2$:

$$F_{\text{drag}} = \frac{1}{2}(5 \times 10^{-13})(7613)^2(2.2)(0.034) \approx 1.1 \times 10^{-6} \text{ N}, \quad \tau_{\text{aero}} \approx F_{\text{drag}} r_{cp} \approx 2.2 \times 10^{-8} \text{ N m.} \quad (6)$$

Density varies by an order of magnitude with solar activity, so ρ is treated as a tunable parameter in the simulation.

4 Solar Radiation Pressure (SRP) Torque

Physical origin

Sunlight carries momentum; photons striking the spacecraft exert a small pressure. As with drag, an offset between the centre of pressure of the illuminated surface and the CoM converts this force into a torque.

Derivation

The solar radiation pressure at Earth's distance is $P = \Phi/c$, where $\Phi \approx 1361 \text{ W/m}^2$ is the solar flux and c the speed of light, giving $P \approx 4.56 \times 10^{-6} \text{ N m}^{-2}$. The force on an illuminated area A_s with reflectivity q ($0 = \text{perfectly absorbing}$, $1 = \text{perfectly reflecting}$) is

$$\mathbf{F}_{\text{srp}} = -P(1+q)A_s\hat{\mathbf{s}}, \quad (7)$$

with $\hat{\mathbf{s}}$ the Sun-to-spacecraft unit vector. The resulting torque about the CoM, with solar-pressure centre offset $\mathbf{r}_{cps}$, is

$$\boxed{\boldsymbol{\tau}_{\text{srp}} = \mathbf{r}_{cps} \times (\mathbf{R}(\mathbf{q})^\top \mathbf{F}_{\text{srp}})} \quad (8)$$

The torque is present only in sunlight (zero during eclipse), so for a Sun-synchronous orbit it acts as a roughly once-per-orbit modulated disturbance.

Order of magnitude

With $q = 0.6$, $A_s = 0.034 \text{ m}^2$, $r_{cps} = 0.02 \text{ m}$:

$$F_{\text{srp}} = (4.56 \times 10^{-6})(1.6)(0.034) \approx 2.5 \times 10^{-7} \text{ N}, \quad \tau_{\text{srp}} \approx 5.0 \times 10^{-9} \text{ N m.} \quad (9)$$

This is the smallest of the four for our platform, but for missions with large deployed solar panels (large A_s and r_{cps}) it can dominate.

5 Residual Magnetic Torque

Physical origin

The spacecraft has a small residual magnetic dipole $\mathbf{m}$ arising from current loops, permanent magnets in components, and magnetised materials. This dipole interacts with Earth's magnetic field $\mathbf{B}$, producing a torque — the same physics the magnetorquers use deliberately for desaturation, here acting as an unwanted disturbance.

Derivation

The torque on a magnetic dipole $\mathbf{m}$ in an external field $\mathbf{B}$ is

$$\tau_{\text{mag}} = \mathbf{m} \times \mathbf{B} \quad (10)$$

where $\mathbf{m}$ is the body’s residual dipole (body frame, in A m^2) and $\mathbf{B} = \mathbf{R}(\mathbf{q})^\top \mathbf{B}_{\text{inertial}}$ is the local geomagnetic field in the body frame. The field magnitude in LEO is $B \approx 2 \times 10^{-5} \text{ T}$ to $5 \times 10^{-5} \text{ T}$ and reverses roughly twice per orbit as the spacecraft crosses the magnetic equator, so this torque is strongly attitude- and position-dependent.

Order of magnitude

A typical uncompensated CubeSat residual dipole is $m \approx 0.01 \text{ A m}^2$ to 0.05 A m^2 . Taking $m = 0.02 \text{ A m}^2$ and $B = 3 \times 10^{-5} \text{ T}$:

$$\tau_{\text{mag}} \approx (0.02)(3 \times 10^{-5}) \approx 6 \times 10^{-7} \text{ N m}. \quad (11)$$

This is often the *largest* of the environmental disturbances for a small satellite, which is why minimising the residual dipole (magnetic cleanliness) is a standard design driver.

6 Summary and Implications

Table 2 collects the four torques. All are of order 10^{-9} N m to 10^{-6} N m — individually tiny, but *secular*: acting continuously, they steadily accumulate angular momentum in the reaction wheels.

Table 2: Environmental disturbance torques for the 3U reference mission.

Disturbance	Governing equation	Magnitude (N m)
Gravity gradient	$3n^2 \hat{\mathbf{o}} \times (\mathbf{J}\hat{\mathbf{o}})$	$\sim 2 \times 10^{-8}$
Aerodynamic drag	$\mathbf{r}_{cp} \times \mathbf{F}_{\text{drag}}$	$\sim 2 \times 10^{-8}$
Solar radiation press.	$\mathbf{r}_{cps} \times \mathbf{F}_{\text{srp}}$	$\sim 5 \times 10^{-9}$
Residual magnetic	$\mathbf{m} \times \mathbf{B}$	$\sim 6 \times 10^{-7}$

Momentum accumulation and desaturation. Because a reaction-wheel array conserves total angular momentum, any secular disturbance τ_d is absorbed by the wheels, whose stored momentum grows approximately linearly, $h_w(t) \approx \tau_d t$. The wheels saturate at $t_{\text{sat}} = h_{\text{max}}/\tau_d$. For our platform ($h_{\text{max}} \approx 7 \times 10^{-3} \text{ N m s}$) under the combined disturbances ($\sim 7 \times 10^{-7} \text{ N m}$ worst case), t_{sat} is of order a few hours, after which the magnetorquers must remove the accumulated momentum by applying a commanded $\boldsymbol{\tau} = \mathbf{m}_{\text{cmd}} \times \mathbf{B}$ against Earth’s field. This is the physical justification for including magnetorquers in the ADCS architecture.

Modelling scope. The simulation implements the gravity-gradient and aerodynamic-drag torques directly (Eqs. 2 and 5), with all parameters exposed in the configuration file. Solar radiation pressure and magnetic torque are documented here for completeness and can be added as further terms in the same disturbance module; each reduces to a body-frame cross product of the form already implemented.